

Computer Vision

Organizational Remarks

Lectures: Kira Maag
Exercises: Arnold Brosch & Ben Hamscher

Visual Analytics
Computer Science Department
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Summer Semester 2026

Educational Objectives

- ▶ Established by splitting the „Deep Learning“ lecture held until summer 2022
 - ▶ Deep learning: Focus on machine learning with deep neural networks
 - ▶ Computer Vision: Focus on understanding how information of the 3D world is encoded in digital images, and can be decoded with computers - also (but not only) using Deep Learning
- ▶ Prerequisites: Linear algebra, ML/DL basics, Programming
- ▶ 5 credit points = ~ 150 working hours

From practice to applications:

- ▶ Introduction to computer vision concepts and architectures
- ▶ Implementation of models using Python and Jupyter notebooks

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 - ▶ Theory and python implementations

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- ▶ Level 3 consolidation via involving **you** into the exercises
 - ▶ Theory and python implementations

Exam:

- ▶ Written or oral exam
- ▶ Everyone is admitted to the exam
- ▶ Make sure to master both the exercises and the lecture materials to be successful!

The Team



Kira

Lecture; Q&A

`kira.maag@hhu.de`



Arnold

Exercises; Q&A

`arnold.brosch@hhu.de`



Ben

Exercises; Q&A

`ben.hamscher@hhu.de`

- ▶ Office hours by appointment
- ▶ Ideally, questions asked during the lecture & exercises; they are very welcome any time; there are **no** stupid questions

Our Background



Kira

Junior Professorship
Research: Safe
& secure DL



Arnold

M.Sc. in Mathematics
Research: Uncertainty
in DL models



Ben

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Research: Robustness
of DL models

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► What is **your** background?

Lecture:

- ▶ Wednesdays, 10:30-12:00, 6G

Exercises:

- ▶ Exercise 1: Monday, 12:30-14:00, 2513.U1.22
 - ▶ Exercise 2: Wednesdays, 12:30-14:00, 6C
- Exercises start in two weeks (27./29. April)

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Communication:

- ▶ We use Ilias for general information
- ▶ Ask questions and discuss on rocket chat:
#2026-sose-computervision

Exercises:

- ▶ The exercises feature optional exercise sheets to strengthen understanding of key concepts with theory tasks
- ▶ Further, 3 practical projects contain implementations and applications of learned concepts and will be discussed in the exercise classes
- ▶ The solutions are presented in the exercises (but not uploaded)
- ▶ The first sheet with a Maths Recap can be found in ILIAS.

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Project Bonus:

- ▶ Submitting the practical projects on time with a passing solution ($> 50\%$ of points) grants bonus points in the exam:
 - ▶ 2 projects grant one grade level (e.g. $3.0 \rightarrow 2.7$)
 - ▶ 3 projects grant two grade levels (e.g. $3.0 \rightarrow 2.3$)
 - ▶ If your exam grade is below passing, bonus points are not applied
- ▶ Form groups of two or three students each

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Do **you** have questions?

Content of the Lecture

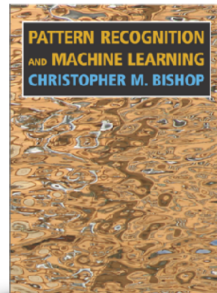
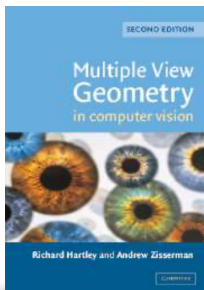
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- ▶ Local Features
- ▶ Epipolar geometry
- ▶ Structure from motion: Multi-view
- ▶ Stereo reconstruction
- ▶ Surface extraction
- ▶ Recognition
- ▶ Differential rendering
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- ▶ **Note:** lecture is continuously under development; slight changes may happen
- ▶ *A big thank to Timo Dickscheid, on whose lecture this one is based*

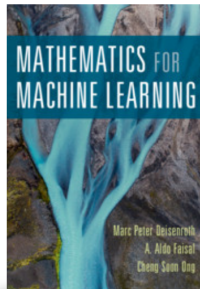
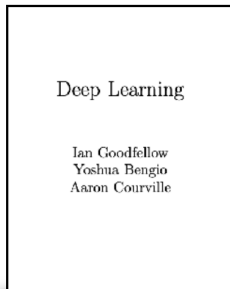
References



- ▶ Good general overview: available from Rick Szeliski
- ▶ About 3D No Machine learning!: available at Oxford
- ▶ Machine learning & graphical models: available at microsoft.com

Note: You can do the course without these references!

References



- ▶ Good general reference on DL: available from deeplearningbook.org
- ▶ Good reference for related mathematics: available at mml-book.github.io